

Yuzheng Hu

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EDUCATION

University of Illinois at Urbana-Champaign

Urbana, IL, USA

Ph.D. in Computer Science

Aug. 2021 – May. 2026 (expected)

- **Advisor:** Han Zhao

Peking University

Beijing, China

B.S. in Mathematics (Math and Applied Math, Honor Track)

Sep. 2017 – Jul. 2021

- **Advisors:** Wei Lin and Liwei Wang

RESEARCH INTERESTS

I develop the **science of data**—a perspective that views data as an *optimizable, expandable* and *predictable* component of AI systems. To this end, I work on **data attribution**, **synthetic data**, and **privacy**.

PUBLICATIONS

★ stands for equal contribution or alphabetical order

- [P1] A Snapshot of Influence: A Local Data Attribution Framework for Online Reinforcement Learning 📄 🔗
Yuzheng Hu★, Fan Wu★, Haotian Ye, David Forsyth, James Zou, Nan Jiang, Jiaqi W. Ma, Han Zhao
The 39th Annual Conference on Neural Information Processing Systems (NeurIPS 2025, Oral)
- [P2] Taming Hyperparameter Sensitivity in Data Attribution: Practical Selection Without Costly Retraining 📄
Weiyi Wang, Junwei Deng, **Yuzheng Hu**, Shiyuan Zhang, Xirui Jiang, Runting Zhang, Han Zhao, Jiaqi W. Ma
The 39th Annual Conference on Neural Information Processing Systems (NeurIPS 2025)
- [P3] MergeBench: A Benchmark for Merging Domain-Specialized LLMs 📄 🔗
Yifei He, Siqi Zeng, **Yuzheng Hu**, Rui Yang, Tong Zhang, Han Zhao
The 39th Annual Conference on Neural Information Processing Systems (NeurIPS 2025, Datasets and Benchmarks Track)
- [P4] Empirical Privacy Variance 📄 🔗
Yuzheng Hu★, Fan Wu★, Ruicheng Xian, Yuhang Liu, Lydia Zakyntinou, Pritish Kamath, Chiyan Zhang, David Forsyth
The 42nd International Conference on Machine Learning (ICML 2025)
- [P5] An Improved Autoregressive Evaluation Paradigm for Large Language Models 📄
Jipeng Zhang★, Rui Pan★, **Yuzheng Hu**★, Kashun Shum★, Guanyu Yao, Xiang Liu, Renjie Pi, Hanze Dong, Shizhe Diao, Yong Lin, Han Zhao, Tong Zhang
ACM Transactions on Intelligent Systems and Technology (ACM TIST 2025)
- [P6] Most Influential Subset Selection: Challenges, Promises, and Beyond 📄 🔗
Yuzheng Hu, Pingbang Hu, Han Zhao, Jiaqi W. Ma
The 38th Annual Conference on Neural Information Processing Systems (NeurIPS 2024)
- [P7] Localize-and-Stitch: Efficient Model Merging via Sparse Task Arithmetic 📄 🔗
Yifei He, **Yuzheng Hu**, Yong Lin, Tong Zhang, Han Zhao
Transactions on Machine Learning Research (TMLR 2024)
- [P8] SoK: Privacy-Preserving Data Synthesis 📄 🌐
Yuzheng Hu★, Fan Wu★, Qinbin Li, Yunhui Long, Gonzalo Munilla Garrido, Chang Ge, Bolin Ding, David Forsyth, Bo Li, Dawn Song
The 45th IEEE Symposium on Security and Privacy (S&P 2024)

- [P9] HPL-ViT: A Unified Perception Framework for Heterogeneous Parallel LiDARs in V2V 
Yuhang Liu, Boyi Sun, Yuke Li, **Yuzheng Hu**, Fei-Yue Wang
The 41st IEEE International Conference on Robotics and Automation (ICRA 2024)
- [P10] Revisiting Scalarization in Multi-Task Learning: A Theoretical Perspective 
Yuzheng Hu, Ruicheng Xian, Qilong Wu, Qiuling Fan, Lang Yin, Han Zhao
The 37th Annual Conference on Neural Information Processing Systems (NeurIPS 2023)
- [P11] Understanding the Impact of Adversarial Robustness on Accuracy Disparity  
Yuzheng Hu, Fan Wu, Hongyang Zhang, Han Zhao
The 40th International Conference on Machine Learning (ICML 2023)
- [P12] Actor-critic is implicitly biased towards high entropy optimal policies 
Yuzheng Hu*, Ziwei Ji*, Matus Telgarsky*
The 10th International Conference on Learning Representations (ICLR 2022)
- [P13] Towards Understanding the Data Dependency of Mixup-style Training  
Muthu Chidambaram, Xiang Wang, **Yuzheng Hu**, Chenwei Wu, Rong Ge
*The 10th International Conference on Learning Representations (ICLR 2022, **Spotlight**)*
- [P14] Don't Waste Your Bits! Squeeze Activations and Gradients for Deep Neural Networks via TinyScript 
Fangcheng Fu, **Yuzheng Hu**, Yihan He, Jiawei Jiang, Yingxia Shao, Ce Zhang, Bin Cui
The 37th International Conference on Machine Learning (ICML 2020)

- MANUSCRIPTS [M1] ACTG-ARL: Differentially Private Conditional Text Generation with RL-Boosted Control  
Yuzheng Hu, Ryan McKenna, Da Yu, Shanshan Wu, Han Zhao, Zheng Xu, Peter Kairouz
- [M2] A Survey of Data Attribution: Methods, Applications, and Evaluation in the Era of Generative AI 
Junwei Deng*, **Yuzheng Hu***, Pingbang Hu*, Ting-Wei Li*, Shixuan Liu*, Jiachen T. Wang, 21 others, Jiaqi W. Ma
- [M3] Humanity's Last Exam (arXiv:2501.14249)   
Dataset contributor. Authored 4 of the 2,500 questions for the benchmark, with one being selected as a prize-winning entry (**top 550 world-wide**)

INVITED TALKS **Differentially Private Conditional Text Generation with RL-Boosted Control**

- Google Athena seminar Sep. 2025
- Google ML Privacy seminar (Host: Natalia Ponomareva) Sep. 2025

Empirical Privacy Variance

- Google ML Privacy seminar (Host: Matthew Jagielski) Mar. 2025

INTERNSHIP EXPERIENCE

Google Research

Research Intern, Hosts: Zheng Xu, Ryan McKenna, Da Yu; Manager: Peter Kairouz

Mountain View, CA, US

May. 2025 – Nov. 2025

- Developed a hybrid approach for DP synthetic text generation that advances the state of the art [M1]

Jane Street Group, LLC

Quantitative Trading Intern

New York, NY, US

May. 2024 – July. 2024

- Ranked 2/200 (1/150 among interns) in the Tenth Annual Figgie World Championships
- Participated in mock trading games and worked on two desk projects (options and equities)

Damo Academy, Alibaba Group (U.S.) Inc.

Research Intern, Mentor: Bolin Ding

Bellevue, WA, US

May. 2022 – Aug. 2022

	◦ Worked on the intersection of privacy and law ◦ Wrote a systematization of knowledge paper on privacy-preserving data synthesis [P8]	
RESEARCH EXPERIENCE	Simons Institute for the Theory of Computing  <i>Visiting Graduate Student, Program: Modern Paradigms in Generalization</i> Berkeley, CA, US Aug. 2024 – Dec. 2024 ◦ Quantified the discrepancy between differential privacy and memorization in LLMs [P4], with Pritish Kamath and Chiyuan Zhang University of Illinois at Urbana-Champaign <i>Graduate Research Assistant, Advisor: Han Zhao</i> Urbana, IL, US Feb. 2022 – now ◦ Thesis research: data attribution [P1] [P2] [P6] [M2], with Jiaqi Ma ◦ Other works: model merging [P3] [P7]; representational limitations of scalarization in multi-task learning [P10]; trade-off between robustness and fairness [P11] University of Illinois at Urbana-Champaign <i>Graduate Research Assistant, Advisor: Matus Telgarsky</i> Urbana, IL, US Aug. 2021 – Dec. 2021 ◦ Proved that actor-critic converges to high entropy optimal policy without explicit regularization [P12] Duke University <i>Undergraduate Research Assistant, Advisor: Rong Ge</i> Remote Jun. 2020 – Feb. 2021 ◦ Demonstrated the data-dependency of mixup [P13] School of Intelligence Science and Technology, Peking University <i>Undergraduate Research Assistant, Advisor: Liwei Wang</i> Beijing, China Aug. 2019 – Jun. 2020 ◦ Identified two non-local properties of the loss landscape of neural networks School of Computer Science, Peking University <i>Undergraduate Research Assistant, Advisor: Bin Cui</i> Beijing, China Aug. 2018 – Aug. 2019 ◦ Introduced a distribution-aware, non-uniform quantization algorithm for multilayer perceptrons [P14]	
TEACHING EXPERIENCE	UIUC CS 591 – Machine Learning Reading Group Spring 2024, Fall 2024 UIUC CS 598 – Transfer Learning Spring 2023 UIUC CS 498 – Trustworthy Machine Learning Fall 2022	
SERVICES	Student Co-Organizer of the Machine Learning Seminar at UIUC  2024 – now Program Committee: TPDP (2025) , ACL L2M2 (2025)  Reviewer: NeurIPS (2022-2025), ICML (2023-2025), ICLR (2024-2026), AISTATS (2024), ICCV (2025)	
HONORS AND AWARDS	NeurIPS 2025 Oral (Top 0.3%, 77/21,575) 2025 NeurIPS 2025 Top Reviewer (Top 8%)  2025 ICML 2025 Outstanding Reviewer (Top 2%)  2025 Prize-Winning Question in Humanity's Last Exam (Top 1%, 550/70,000+, \$500)  2025 NeurIPS 2023 Top Reviewer (Top 10%)  2023 Saburo Muroga Endowed Fellowship (Top 4% in UIUC, 4/100+)  2021-2022 Exceptional Award for Academic Innovation (Top 1% in Peking University) 2020 Elite Undergraduate Training Program of Applied Math (Top 15% in SMS)  2019-2021 Invited Participant, China TST for the 58th International Mathematical Olympiad (IMO) 2017 Gold Medal, 32nd Chinese Mathematical Olympiad (CMO) (Top 60 in China) 2016 Scored 771/810 in Jr. High Entrance Exam (Top 0.01% in Guangzhou, 10/100,000+) 2014	

TRAVEL GRANTS	TPDP Student Travel Funding (\$500)	2025
	IEEE S&P Student Travel Grants Award (\$1000)	2024
	NeurIPS 2023 Scholar Award (\$2000)	2023
SKILLS	Programming Frameworks & Packages	Python, Shell Script, $\text{\LaTeX}$, MATLAB PyTorch, HuggingFace